







Sign in

Username *
 firsttimeuser

Password *
 ••••••••

☐ Remember me

SIGN IN

EECS Dept., University of Missouri 2024.

- On successful password authentication, the user is taken to the “About” page.

Welcome firsttimeuser <

- Dashboard
- Account
- Manage Users
- Manage Edge Devices
- Add Nodes
- Manage Policies
- Plan Missions
- Mission Executions
- Deploy Honeypots
- Attack Metrics
- Blacklist
- About Arculus
- Sign Out

ARCULUS GROUND CONTROL CLIENT

Zero Trust Mode:

Do not enforce Zero Trust Security



About Arculus

Arculus is a cutting-edge project being developed by the Cyber Education, Research and Infrastructure (CERI) Center at the University of Missouri - Columbia. It represents a significant advancement in cyber security solutions, focusing on a Low Overhead Zero Trust Solution in a Tactical Edge Network setting. Key features of Arculus include Microsegmentation and Active Defense using Honeypots. The Microsegmentation approach effectively prevents lateral movement of threat agents.

- The “Manage Users” tab of the Arculus sidebar takes the user to the “User Management Dashboard” where (s)he (only mission creators) can view existing users, create new users, configure and edit their roles and domains, or delete the users. For a demo purpose, we created a user named “testuser” configured with a “Mission Supervisor” role.

Welcome firsttimeuser <

- Dashboard
- Account
- Manage Users
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- Add Nodes
- Manage Policies
- Plan Missions

ARCULUS GROUND CONTROL CLIENT

Zero Trust Mode:

Enforce Risk-based Zero Trust Security

User Management Dashboard

User Name	E-mail ID	Role	Domains	Action	Action
firsttimeuser	firsttimeuser@missouri.edu	Mission Creator	Explosive Ordnance Reconnaissance Covert Operations Support Structural Integrity Assessment Crisis Response and Restoration		

ADD USER

Welcome firsttimeuser <

ARCULUS GROUND CONTROL CLIENT

Zero Trust Mode: Enforce Risk-based Zero Trust Security

User Management Dashboard

User Name	E-mail ID	Role	Domains	Action	Action
firsttimeuser	firsttimeuser@missouri.edu	Mission Creator	Explosive Ordnance Reconnaissance Covert Operations Support Structural Integrity Assessment Crisis Response and Restoration		
testuser	tester@gmail.com	Mission Supervisor	Covert Operations Support	EDIT	DELETE

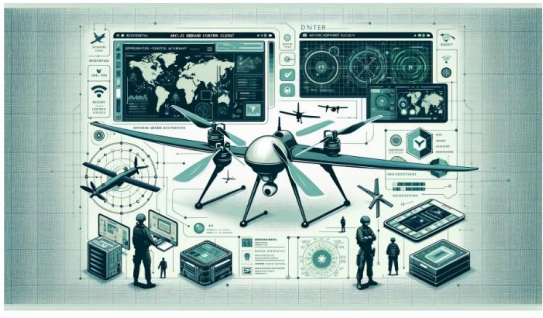
ADD USER

5. On signing in as testuser (a Mission Supervisor), we can observe that there are limited dashboards (s)he can navigate to, demonstrating a robust Role-based Access Control for users' interaction with the UI.

Welcome testuser <

ARCULUS GROUND CONTROL CLIENT

Zero Trust Mode: Enforce Risk-based Zero Trust Security



About Arculus

Arculus is a cutting-edge project being developed by the **Cyber Education, Research and Infrastructure (CERI) Center at the University of Missouri - Columbia**. It represents a significant advancement in cyber security solutions, focusing on a **Low Overhead Zero Trust Solution in a Tactical Edge Network setting**. Key features of Arculus include **Microsegmentation** and **Active Defense using Honeypots**. The Microsegmentation approach effectively prevents lateral movement of threat agents.

6. The more important functionality of device management begins with the “Add Nodes” tab of the sidebar. This page generates custom scripts for new devices to join the Arculus device cluster. On entering the name we want to give the device that joins the cluster, a script is created with the parameter. This script can be copied to clipboard and be run on the device we want to add to the cluster.

Welcome firsttimeuser < **ARCULUS GROUND CONTROL CLIE...** **Zero Trust Mode:** Enforce Risk-based Zero Trust Security

Dashboard
Account
Manage Users
Manage Edge Devices
Add Nodes
Manage Policies
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Mission Executions
Deploy Honey pots
Attack Metrics
Blacklist
About Arculus
Sign Out




Download both the Arculus Join Request Wizard and the Honeypot Wizard here.

ARCULUS JOIN REQUEST WIZARD
ARCULUS HONEYPOT WIZARD

Or pull the scripts through CLI to install.

Enter the node name you want to join to the cluster with (Would not work without this):
surveillancedrone

Join through CLI
wait
chmod +x joinClusterWizard.sh
chmod +x honeypotWiz.py
./joinClusterWizard.sh 52.23.225.56 172.31.26.207 surveillancedrone

Copied to Clipboard

- The automated script puts in a join request to the Arculus backend. This request pops up on the Device Management Dashboard of the Arculus UI under the “Cluster Join Requests” section.

```
ubuntu@ip-172-31-47-238:~$ sudo su
root@ip-172-31-47-238:/home/ubuntu# curl -o joinClusterWizard.sh http://52.23.225.56:3001/tools/downloadJoinWiz & curl -o honeypotWiz.py http://52.23.225.56:3001/tools/downloadHoneyWiz &
wait
chmod +x joinClusterWizard.sh
chmod +x honeypotWiz.py
./joinClusterWizard.sh 52.23.225.56 172.31.26.207 surveillancedrone
```

```
0 upgraded, 0 newly installed, 0 to remove and 26 not upgraded.
Cluster join request successful.
Opening websocket to get join status...
ws://52.23.225.56:3002/getJoinStatus?nodeName=surveillancedrone
Received message: Request submitted. Please hang on until an administrator approves your request.
```

Welcome firsttimeuser < **ARCULUS GROUND CONT...** **Zero Trust Mode:** Enforce Risk-based

Device Management Dashboard

Cluster Join Requests	Configured Devices
 34.230.83.210	More Devices in the Cluster  controller

- Super users (Mission Creators) can view these requests and choose to either approve or decline the request. On approval, the K3s token of the Arculus backend is securely shared with the new device, with which is joins the cluster.

×

Requesting Device Information

Name	IP Address
surveillancedrone	34.230.83.210

✓ APPROVE

✗ DECLINE

```

Received message: Request submitted. Please hang on until an administrator approves your request.
Join Successful. Proceeding further.
Token received: K102cbcaebf4feecbe1860cf21abe1a5e3d4fba215fa092b3684df4724f8eb56e22::server:12e8262b81ea34d0
sudo: /usr/local/bin/k3s-agent-uninstall.sh: command not found
Failed to stop k3s.service: Unit k3s.service not loaded.
Failed to disable unit: Unit file k3s.service does not exist.
[INFO] Finding release for channel stable
[INFO] Using v1.29.6+k3s2 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.29.6+k3s2/sha256sum-amd64
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.29.6+k3s2/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubect1 symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Creating /usr/local/bin/ctr symlink to k3s
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-agent-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s-agent.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s-agent.service
[INFO] systemd: Enabling k3s-agent unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s-agent.service → /etc/systemd/system/k3s-agent.service
[INFO] systemd: Starting k3s-agent

```

- Once the new device sets up all the required dependencies on itself and joins the cluster, it shows up on the Device Management Dashboard under the “More Devices in the Cluster” section.

Welcome firsttimeuser <

ARCULUS GROUND CONT...

▶

Zero Trust Mode: Enforce Risk-based

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Dashboard

Account

Manage Users

Manage Edge Devices

Add Nodes

Manage Policies

Plan Missions

Mission Executions

Deploy Honey pots

Device Management Dashboard

Cluster Join Requests

Configured Devices

More Devices in the Cluster

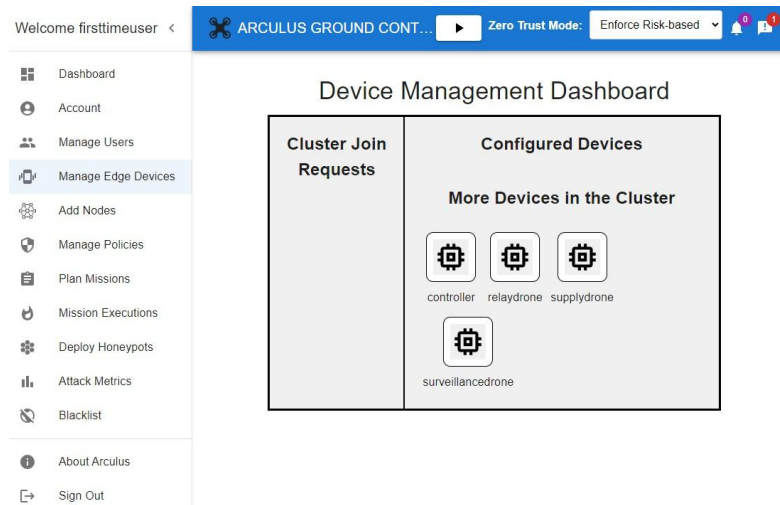
🔧

controller

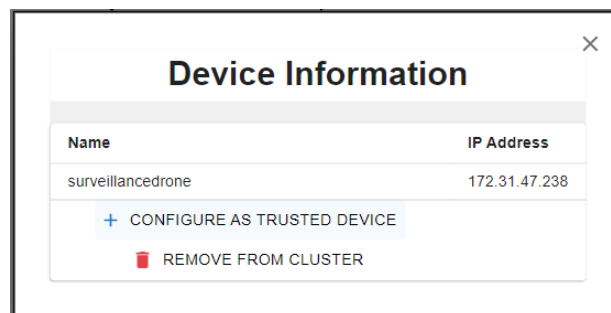
🔧

surveillancedrone

- As we would need a surveillance drone, a supply drone, a relay drone and a controller, add two more devices to the cluster using the same procedure as above.



11. On clicking on an device from the “More Devices” section, we can configure them further as trusted devices to attain virtualized behavior by deploying containers on them or remove the devices from the cluster.



12. By configuring the devices as trusted devices, we can deploy them as certain types of drones/rovers/devices with respective functionalities, and adjust the allowed operations for the devices, which dictate the ingress and egress rules for the devices.

Add Trusted Device

Device Name *
surveillancedrone

IP Address *
172.31.47.238

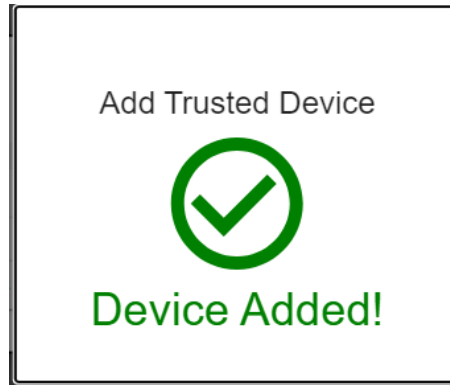
The device will be assigned a local IP address after setup and will no longer be referenced by the public IP address.

Device Type *
Video Capture Drone

Allowed Operations
send_video × send_sensordata ×
receive_command ×

Possible Ingress Rules *
5025/TCP

Egress Rules *
5005/UDP
5035/TCP



13. Configure all four devices required for the mission execution with their respective drone types and capabilities as shown in the configuration screenshots below.

Welcome firsttimeuser < **ARCULUS GROUND CONT...** Zero Trust Mode: Enforce Risk-based

Device Management Dashboard

Cluster Join Requests

Configured Devices

surveillancedrone

controller

relaydrone

supplydrone

More Devices in the Cluster

Configured Device Information

Name	IP Address	Device Type	Allowed Operations
surveillancedrone	10.42.1.3	Video Capture Drone	send_video send_sensordata send_posdata receive_command

UPDATE DEVICE CAPABILITIES

Configured Device Information

Name	IP Address	Device Type	Allowed Operations
controller	10.42.0.9	Video Analytic Controller	send_command receive_posdata receive_sensordata receive_video

UPDATE DEVICE CAPABILITIES

Configured Device Information

Name	IP Address	Device Type	Allowed Operations
relaydrone	10.42.3.3	Communication Relay Drone	send_command send_video send_sensordata send_posdata receive_posdata receive_sensordata receive_video receive_command

UPDATE DEVICE CAPABILITIES

Configured Device Information

Name	IP Address	Device Type	Allowed Operations
supplydrone	10.42.2.3	Freight Drone	send_sensordata send_posdata receive_command

UPDATE DEVICE CAPABILITIES

IV. Deploying, Managing and Monitoring Honeypots

1. The Honeypot dashboard of the Arculus web interface allows for the seamless deployment of multiple types of honeypots. The target device and the honeypot can be selected based on the type of vulnerabilities to be simulated for active defense using the capabilities of the CHN Server.

The screenshot shows the 'Honeypot Dashboard' in the Arculus web interface. The left sidebar contains navigation links: Dashboard, Account, Manage Users, Manage Edge Devices, Add Nodes, Manage Policies, Plan Missions, Mission Executions, Deploy Honeypots (selected), Attack Metrics, Blacklist, About Arculus, and Sign Out. The main content area has a header with 'Welcome firsttimeuser', 'ARCULUS GROUND CONTROL CLI...', 'Zero Trust Mode: Enforce Risk-based Zero Trust Secur...', and notification icons. Below the header, there are tabs for 'Deployed HoneyPots' and 'Deploy HoneyPots'. The 'Deploy New Honeypot' section includes a 'Device IP Address' dropdown set to '172.31.45.90 (relaydrone)', a 'Honeypot Type' dropdown set to 'Cowrie', a 'Honeypot Description' text area with details about the Cowrie honeypot, and a 'Trap Port Mapping' field set to '2222, 2222, 23, 2223'. A 'DEPLOY HONEYPOT' button is at the bottom. The footer indicates 'Powered by CommunityHoneyNet (CHN)'.

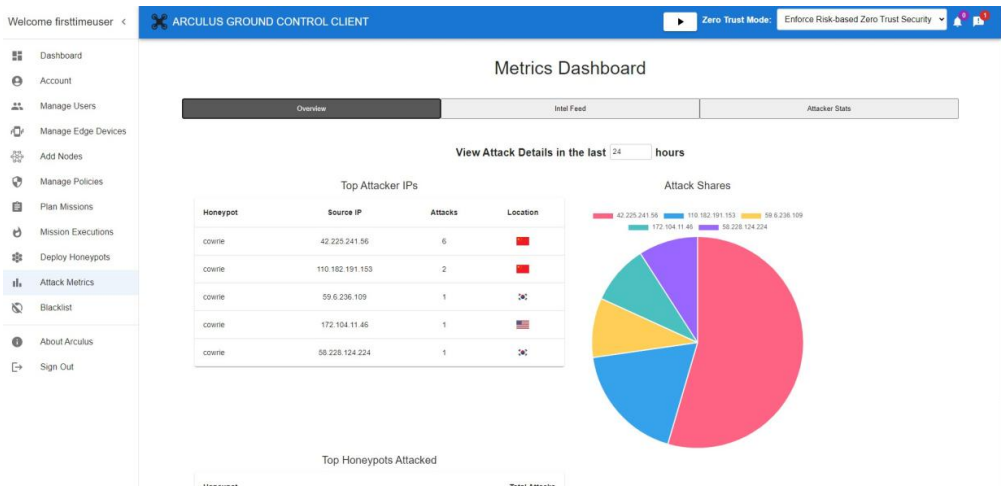
2. The deployed honeypots can be viewed or deleted on the “Deployed Honeypots” of the Honeypot dashboard.

The screenshot shows the 'Honeypot Dashboard' with the 'Deployed HoneyPots' tab selected. The table displays the following data:

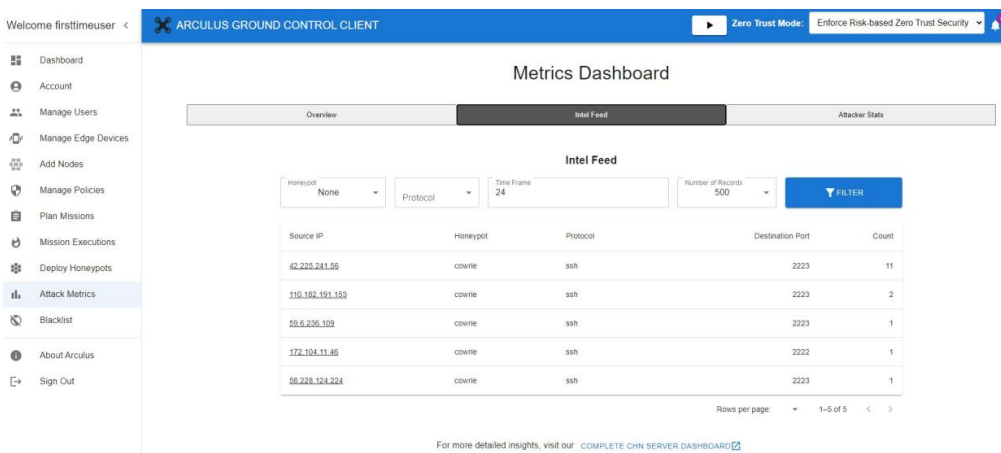
IP Address	Honeypot Type	Actions
172.31.45.90	Cowrie	View Delete

The footer indicates 'Powered by CommunityHoneyNet (CHN)'.

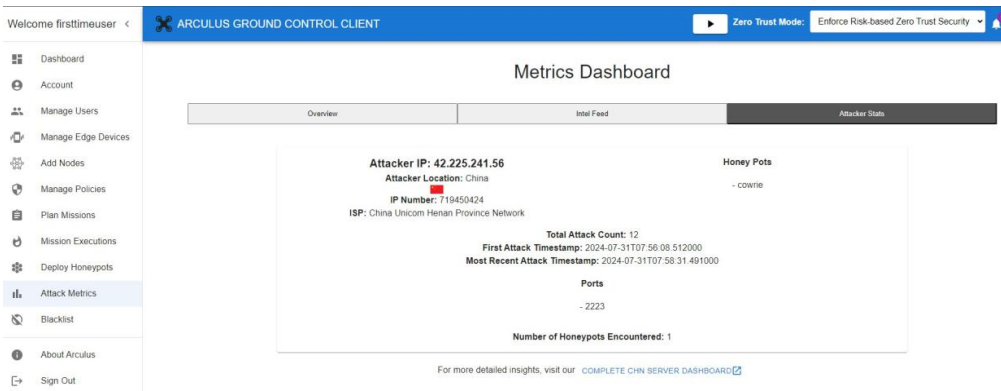
3. On the Metrics Dashboard, we can see the overview of all the attackers and their share of the total malicious activity on our sensors.



4. The Intel Feed gives deep and technical insights into what types of attacks using which protocols and ports were executed on the cluster. This gives us information on the strategies attackers are taking as a pattern to establish attack signatures.



5. We can also view the “Attacker Stats” tab where a detailed information of the source of attacks and the total count of the attack attempts can be seen. These dashboards provide a lot of threat intelligence which is also backed up on the CHN Server for detailed analysis.



V. Planning and Execution of Drone Missions on the Arculus Testbed

1. The Mission Planning Dashboard allows for configuring drone missions by specifying which devices to use, what the context (criticality) of the mission is, who the supervisors and viewers of the mission are, and logistics information. Configure these options to tune the criticality of the mission dictating the zero trust levels. The device fields are automatically populated based on the availability of the suitable types of devices.

Welcome firsttimeuser < **ARCULUS GROUND CONTROL CLIENT** Zero Trust Mode: Enforce Risk-based Zero Trust Security

Mission Planning Dashboard

Select Mission Type | Plan Mission

Plan Mission

Selected Mission Type: Stealthy Reconnaissance and Resupply

Mission Location: Battlefield 1: Palojo Forest | **Video-Analytic Route Planner (Ground Control):** controller

Video Collection Surveillance Drone: surveillancedrone | **Supply Delivery Drone:** supplydrone

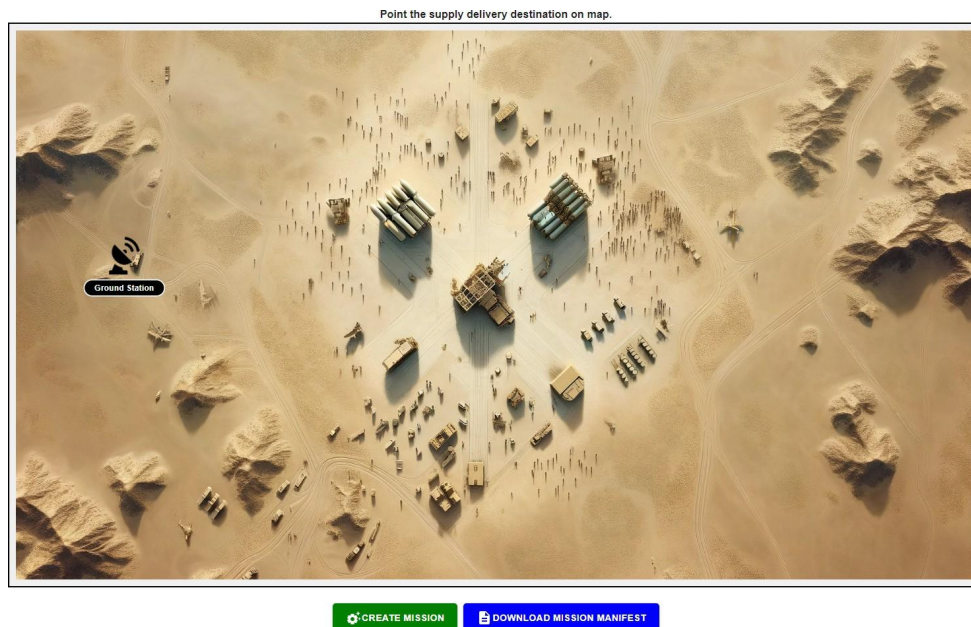
Communication Relay Drone: relaydrone | **Asset Criticality:** Low

Life Threat: Low | **Data Sensitivity:** Low

Strategic Importance: Low | **Supervisors:** testuser

Viewers:

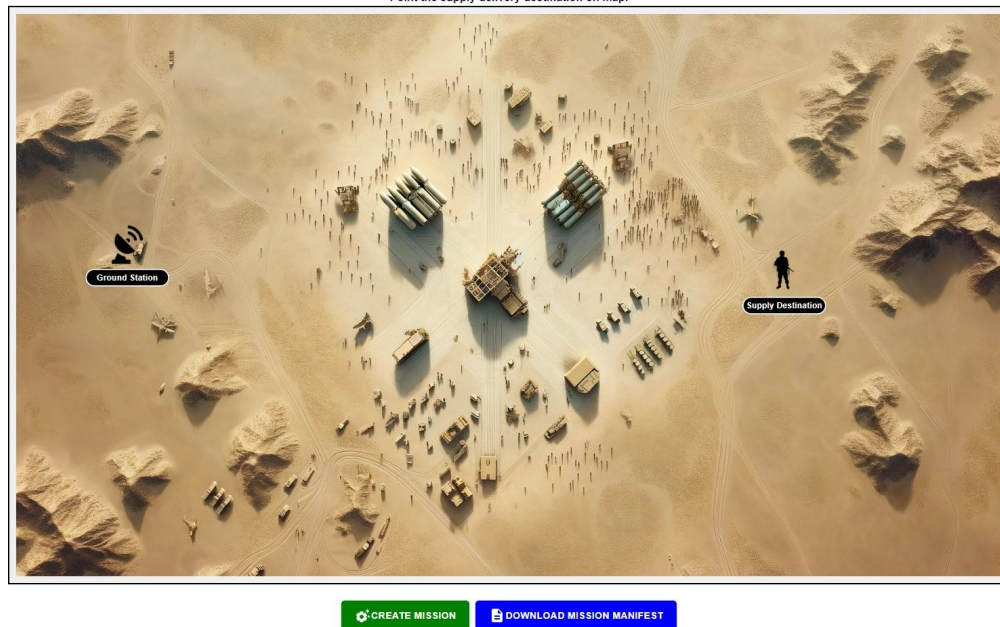
2. The page also provides maps of locations to choose from. On these maps, users can pick a destination point for the delivery of supplies (here, the case of Stealthy Reconnaissance and Resupply mission). Try creating the mission without pointing a destination on the map, and the mission creation will not go ahead asking the user to place a destination on the map.



Please point the destination on the map!

OK

- Click anywhere on the map to set the supply destination to that location.



- Try updating the capabilities and remove one of the privileges specified in the device configuration step. The mission creation won't go ahead because, inability of any of the devices to perform their respective functions will cause the mission to fail for sure. This validation mechanism is a proactive approach to missions whose failure cannot be afforded.

Edit Trusted Device

Device Name *
surveillancedrone

IP Address *
10.42.1.3

Allowed Operations

- send_video
- send_sensordata
- receive_command
- send_posdata

Possible Ingress Rules *

- 5005/UDP
- 5015/TCP
- 5035/TCP

Egress Rules *

- 5025/TCP

UPDATE TRUSTED DEVICE

CANCEL

Insufficient Privileges for Some Devices

Mission Execution might fail! Please provide sufficient capabilities for necessary ingress and egress.

Device	Privileges
controller	☒ send_command ☒ receive_video ☒ receive_posdata
surveillancedrone	☒ send_video ☒ receive_command ☒ send_posdata
supplydrone	☒ receive_command ☒ send_posdata
relaydrone	☒ receive_command ☒ send_command ☒ send_video ☒ send_posdata ☒ receive_posdata ☒ receive_video

OK

-
- The screenshot shows a Windows File Explorer window with the title bar 'Save As'. The address bar shows the path 'Desktop > Mission Manifests'. The left sidebar shows the 'Desktop' folder selected. The main area displays 'No items match your search.' with columns for 'Name', 'Date modified', 'Type', and 'Size'. At the bottom, the 'File name' field contains 'mission_manifest_2024-07-31T07-18-11.853Z.mconf' and the 'Save as type' is set to 'MCONF File (*.mconf)'. The 'Save' button is highlighted.

- ```
mission_manifest_2024-07-31T07-1
+

File Edit View

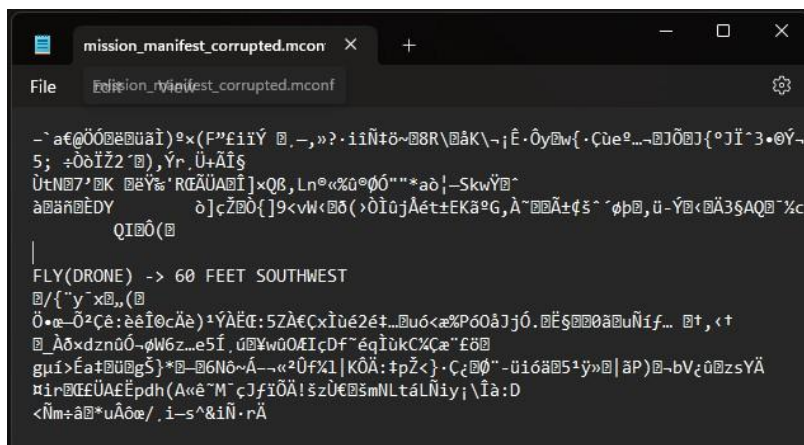
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5; ÷0iZ2`E),Yr.Ü+ÂÎ§
ÛtN7P7K BæY%`RÊÄUABÎI]×QB,Ln«%×0Ü0""*að!-SkwY^
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B_ä0xdzn0Ü·0WZ...e5Î,üB¥w0Ü0iCxDf"éqïùkCXÇæ"Ê0B
guI>Êa+0Ü0g5)*B-B0N0~Ä--«²ÜfX1|K0Ä:~pZ<·ÇzB0"-üi0äB5iÿ>B(äP)B~bVz0Ü0z5YÄ
HirB0ÊÜAEÊpdh(A«Ê"m"çJfï0Ä!szÜ0ÊsmNltäLNiyj\Iä:D
<Nîm:äB*UÄ0æ/.i-s-Ä&iN.rÄ
```

|                         |                                     |                   |
|-------------------------|-------------------------------------|-------------------|
| Missions created by you | Execute using Mission Manifest File | Mission Execution |
|-------------------------|-------------------------------------|-------------------|

[SELECT A DIFFERENT MANIFEST FILE](#) mission\_manifest\_2024-07-31T07-18-11.853Z.mconf [X](#) [UPLOAD FILE](#)

EXECUTE MISSION

7. However, try adding your own text to the manifest file and try to poison it as shown below. The application throws an error message identifying the manifest file to be bad.



## Mission Execution Dashboard

Missions created by you

Execute using Mission Manifest File

Mission Execution

Upload Manifest File for Mission Execution

SELECT A DIFFERENT MANIFEST FILE

mission\_manifest\_corrupted.mconf

UPLOAD FILE

Bad File! Please choose an actual .mconf file.

Mission Configuration:  
Waiting for file upload...

EXECUTE MISSION

8. Before trying to execute a mission, click the “Manage Policies” tab that takes you to the Network Policy Management Dashboard. It can be seen as in the below screenshot, that there are no active network policies and all traffic is denied by default.

ARCULUS GROUND CONTROL CLIENT

Zero Trust Mode: Enforce Risk-based Zero Trust Security

Network Policy Management Dashboard

Active Network Policies

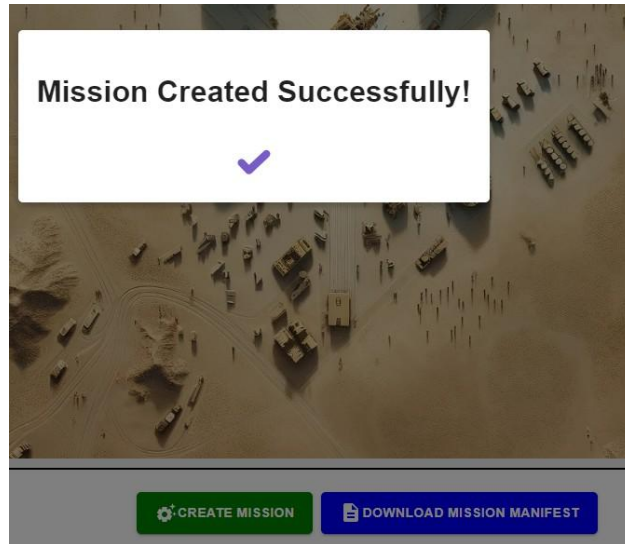
New Network Policy



## No Active Network Policies (Deny By Default)

9. Now, let us create a mission on the mission planning dashboard and execute it using the EXECUTE (▶) button on the “Missions created by me” tab. The mission starts executing within seconds and the application takes us to the “Mission Execution” tab where the live mission and its logs can be monitored.





## Mission Execution Dashboard

| Missions created by you |                                      |            | Execute using Mission Manifest File |         |                     | Mission Execution |         |         |         |        |
|-------------------------|--------------------------------------|------------|-------------------------------------|---------|---------------------|-------------------|---------|---------|---------|--------|
| Mission Location        | Mission Type                         | Creator    | Supervisors                         | Viewers | Creation Time       | Duration          | Execute | Monitor | Status  | Delete |
| Grand Senora Desert     | Stealthy Reconnaissance and Resupply | User admin |                                     |         | 2024-07-31 07:19:27 | 120 sec           | ▶       | 👁       | CREATED | 🗑      |

## Mission Execution Dashboard

Missions created by you

Execute using Mission Manifest File

Mission Execution

Simulate Communication Loss

Simulate GPS Spoofing

Simulate Physical Capture

Simulate Denial of Service

Simulate Low Battery

Simulate Brute Force SSH

The map shows a desert environment with various military assets. A "Ground Station" is marked with a black icon and a "Supply Destination" with a yellow icon. A "Surveillance Drone" is shown in flight, with a green dashed line indicating its path. A text box in the top right corner displays coordinates and status: "Ground Control Station: 282.88, 428.00", "Surveillance Drone: 422.33, 345.67", "Supply Drone: 302.52, 428.54", "Destination: 1300.74, 96.30", and "Status: Active/Ready/Unknown".

Activity Log

Control Remotely

Surveillance Drone

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Abort Mission

- It can be observed that, on navigating to the Network Policy Management Dashboard, new policies are created during the mission execution to facilitate communication for respective tasks. However, as soon as the mission execution finishes/terminates, all the network policies are deleted. This demonstrates the extraordinary and dynamic execution of Task-based Access Control (TBAC) for good privilege management for devices on the network.

